Task Report Day 3

Title: Text Processing (NLP)

TASK-03: Developing a Keyword-Based Search & Ranking Engine

Objective

The objective of this task is to build a Mini Search Engine. In this, we take a raw text file (corpus), clean the text using preprocessing techniques, and then implement a ranking algorithm to find the most relevant sentences based on a user's query.

Tools Used

- VS Code
- Git & GitHub
- MS Word

Libraries Used

- **re:** Used to clean text by removing unwanted characters using patterns.

Concepts Used

- **Preprocessing**: Converting text to lowercase, removing punctuation, and removing stopwords.
- **Tokenization**: Splitting sentences into words (tokens).
- **Jaccard Similarity**: Used to calculate similarity between query and sentences.
- **Ranking**: Sorting sentences based on similarity score.

Errors

1. Punctuation Not Removed Properly

Special characters were still present.

Solution:

I removed the string and imported re regular expression for better results.
Used: re.sub(r'[^\w\s]', '', text)

2. Zero Scores for All Results

Sometimes all sentences were getting score 0.

Reason:
Query words were not matching with sentence words.

Solution:
Checked preprocessing and used meaningful query.

3. Empty Query Input

If the user enters an empty query, the program gives meaningless results.

Solution:
Added a check:

```
If query.strip() == “ ”:
        print("Please enter a valid query")
```

Output

```
PS D:\Qualtech\Daily Tasks\Day3> py -3.13 search-engine.py
Enter your search query: machine learning

Top Results:

Score: 0.22 | Python is the primary language used for data science and machine learning.
Score: 0.10 | Learning to play a musical instrument can improve cognitive function in children.
Score: 0.09 | PyTorch and TensorFlow are the two most popular frameworks for deep learning.
Score: 0.08 | Zero-shot learning allows a model to recognize objects it has never seen before.
Score: 0.08 | Reinforcement Learning from Human Feedback (RLHF) helps align AI with values.
PS D:\Qualtech\Daily Tasks\Day3>
```

```
PS D:\Qualtech\Daily Tasks\Day3> py -3.13 search-engine.py
Enter your search query:
Please enter a valid query
PS D:\Qualtech\Daily Tasks\Day3>
```

Conclusion

This task helped me understand how a basic search engine works. I learned how to clean and process text data, convert it into tokens, and apply similarity algorithms like Jaccard Similarity to rank results.